ORION Paper V: Self-ORION—Failure-Governed Method Evolution without Self-Promotion

Working framework draft

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Abstract

Self-improving research systems face an authority problem: the same process that proposes a method change can often influence the evidence or evaluator used to justify that change. ORION treats method evolution as protected research rather than direct self-editing. Failed, blocked, partial and cannot-check executions are retained as persistent evidence; recurrence is separated from causal responsibility; unresolved issues maintain competing cause hypotheses and discriminator evidence across interventions; invention is gated until ordinary causes and transfer alternatives are challenged; candidate changes execute in isolated, content-addressed environments; and method reuse requires replay, fresh transfer and protected assurance bound to the exact candidate and evidence lineage. The controller has no self-merge primitive and its strongest internal outcome is a recommendation to an external host. Local hostile tests exercise recurrence-not-cause, negative-history retention, replay/fresh-transfer separation, invention readiness and protected-path reward-hacking attacks. These results establish implementation and governance semantics only. A diagnostic glm-5.2 hidden-cause attribution archive scores 21/24 with three residual errors retained; this does not establish transferable self-improvement. Governed self-improvement on fresh development tasks remains externally `CANNOT_CHECK`.

Keywords: self-improving agents; failure attribution; fresh transfer; evaluator integrity; program evolution; AI governance.

1 Problem and scope

A method-evolving research system must answer two different questions: what change should be tried, and who is allowed to conclude that the change is better. Collapsing those questions creates a self-promotion channel. A coding agent may produce an impressive patch, but if it can also rewrite the benchmark, choose only favorable episodes, change protected governance, or certify its own evidence, improvement is not established.

Self-ORION therefore treats development as an evidence-governed research fibre rather than a direct self-edit loop. Its candidate claim is deliberately narrower than "autonomous self-improvement":

> Failures may become method knowledge only after persistent recording, causal discrimination, nearest-work/transfer challenge, isolated challenger execution, replay, fresh transfer and protected assurance; the candidate process cannot grant itself method authority or merge its own change.

The method state M_t from Paper I is therefore not writable merely because a research run failed. A failure first changes the evidence available *about* the method. Only a separately governed transition can change which method is authorized.

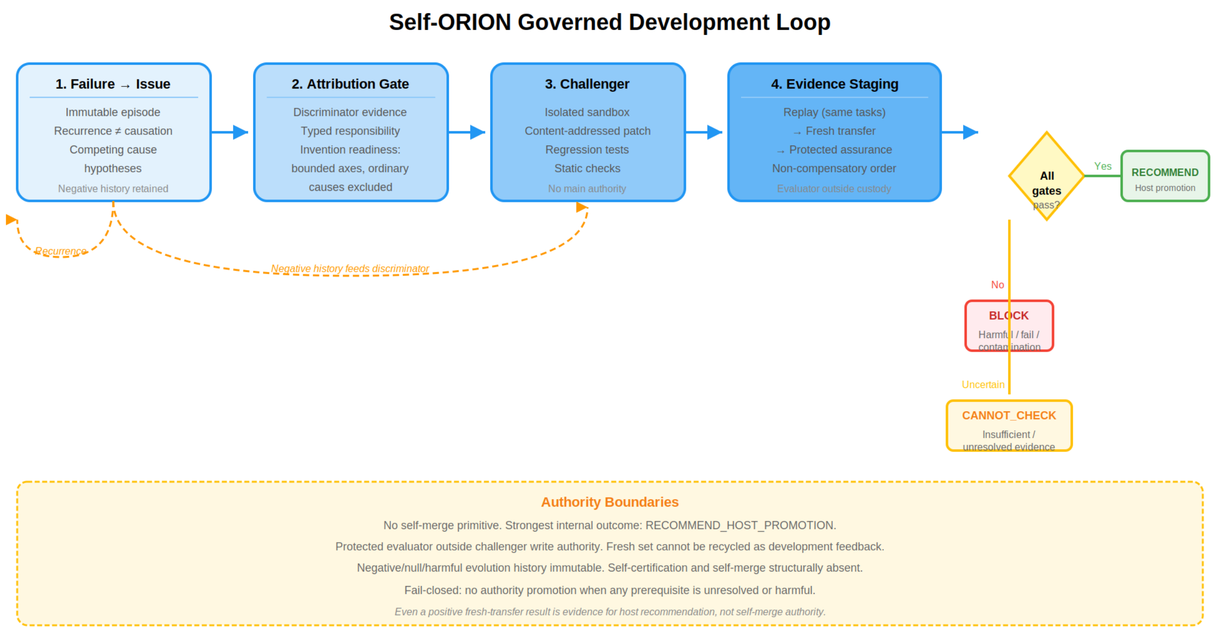

Figure 1: Self-ORION governed development loop. The failure-to-method fibre proceeds through persistent issue recording, causal discrimination, invention-readiness gating, isolated challenger execution, non-compensatory evidence staging (replay → fresh transfer → protected assurance), and a fail-closed authority boundary. The strongest internal outcome is a host recommendation; self-merge and self-certification are structurally absent.

1.1 Candidate deltas

The programme tracks four surviving candidate differences after nearest-work absorption:

1. a failure-to-method fibre that separates observation, recurrence, attribution, lesson/challenger and authority;

2. non-self-promotional evolution in which candidate generation and candidate authorization remain distinct;

3. protected fresh assurance that requires evidence beyond the episodes used to construct the repair;

4. a separate authority layer whose final promotion remains in external host custody.

These are candidate differences, not established novelty or empirical superiority.

2 Failure and null outcomes as persistent knowledge

Self-ORION stores failed, blocked, partial and `CANNOT_CHECK` executions as addressable experience rather than disposable execution noise. A failure episode distinguishes the observed mode, propagated effect, task/variation identity, competing causes, detection evidence, recovery actions, residuals and falsifiers. Negative outcomes remain in history after later success.

This makes failure a potential research input, but not an automatic method lesson. Let e_j denote an immutable episode and $\phi(e_j)$ its observed failure signature. Repeated signatures can justify a recurrence hypothesis,

$$\phi(e_1) \approx \phi(e_2) \approx \cdots \approx \phi(e_n),$$

without justifying the causal statement that one mechanism caused every failure. Recurrence answers "does something structurally similar persist?"; attribution answers "which intervention should change the outcome?"

A valid development record therefore keeps failures, nulls and harmful interventions visible. Otherwise an adaptive system can manufacture an apparently improving history by retaining only successful challengers. Self-ORION treats negative-history completeness as an assurance coordinate rather than an optional debugging convenience.

2.1 Boundary with Paper I

Paper I owns the mechanic-cell representation that can expose a missing or failed coordinate. Paper V begins when execution evidence is interpreted as possible method evidence. The transition from an open mechanic coordinate to a method change is deliberately non-automatic: a missing specification, a provider outage, a data defect and a genuine operator defect require different responses.

3 Causal discrimination and persistent issue state

A development issue persists across multiple investigations and interventions. `DevelopmentIssue.v1` keeps a stable issue identity together with candidate causes, supported causes, discriminator evidence, failure episodes, intervention outcomes and lifecycle transitions. This prevents each repair attempt from becoming a new context that forgets why earlier alternatives failed.

For issue u, let

$$H(u) = \{h_1, h_2, \ldots, h_k\}$$

be competing cause hypotheses. A repair is not licensed merely because one hypothesis is narratively plausible. Self-ORION requires discriminator evidence capable of separating relevant alternatives. Where the discriminator cannot distinguish the hypotheses, the correct state is unresolved rather than an invented causal label.

This ordering is important:

$$\text{observed failure} \to \text{competing causes} \to \text{discriminator} \to \text{attribution} \to \text{candidate intervention.}$$

Moving the intervention before the discriminator encourages post-hoc explanations of whatever the candidate happened to change.

Persistent issue state also makes negative interventions scientifically useful. A harmful or null repair can narrow the supported cause set, expose a missing interaction, or invalidate an earlier lesson. Such episodes remain attached to the issue rather than being deleted once a later challenger succeeds.

4 Nearest-work challenge, challenger archives, and invention readiness

Self-ORION does not interpret “repeated failure” as “invent a new operator.” Before structural invention, the system searches for ordinary causes, known mechanisms, transfer candidates and mature parent-domain treatments. The invention-readiness gate asks whether relevant search axes are bounded-flat, ordinary causes have been excluded to the supported extent, the residual survives distinct variations, and discriminator evidence identifies a missing-structure class rather than a generic score deficit.

Candidate method changes are retained as challengers rather than silently overwriting the incumbent. The archive records candidate identity, generating issue/evidence, attempted intervention, execution result, regression result, fresh-transfer result, protected-assurance outcome and final host disposition. Failed and harmful candidates remain queryable.

This structure absorbs useful mechanisms from evolutionary and self-improving-agent work without inheriting self-authorization. Archives, iterative candidate generation, population search and evaluator-guided optimization may all be valuable proposal mechanisms. They become method authority only through the separate assurance path.

4.1 Method-level challengers and method-space expansion

A method challenger is a candidate structural change linked to documented failure or obstruction records, not a claim that a new method has been invented. The versioned `MethodChallenger.v1` record binds the subject revision, generating failures, competing ordinary causes already challenged, known-method search routes and their inadequacy, assimilated donor mechanisms, the structural edit and its predeclared discriminator. Candidate generation is therefore separate from host adoption. A candidate must traverse

$$\textsc{Diagnose} \to \textsc{Search/Absorb} \to \textsc{Challenger} \to \textsc{IsolatedRun} \to \textsc{Replay} \to \textsc{Fresh} \to \textsc{Protected} \to \textsc{Hos}$$

A replay improvement cannot compensate for harmful fresh transfer, and missing protected evidence blocks disposition. Rejected, null and harmful histories remain append-only and queryable; neither a generator nor a challenger can rewrite the evaluator or authority registry. For example, a structural adaptation that improves its motivating replay but harms a protected family on fresh transfer remains rejected, even if its development score is higher. Known-method search and transfer must be challenged before an invention gap is declared; repeated failure alone never yields `METHOD_INVENTION_READY`. The host alone may adopt or recommend adoption.

This is a P5 governance boundary, not a P10 invention result. P5 can govern whether a method candidate is safely evaluated and presented to a host; P10 would be required to establish whether a learned system can generate valuable new method structures.

4.2 RAKL transfer as prior design knowledge

The historical Shadow programme re-audited mature RAKL method surfaces as prior design knowledge. Fourteen provisional structural dimensions across 59 registered mechanics produced 826 explicit transfer questions; evidence-bound `AnswerRecord` applications moved that registered structural frontier from 826 to 0 with attributable answer/audit closure.

This result is deliberately narrow:

$$\text{specification closure} \neq \text{empirical validity} \neq \text{promotion authority}.$$

It can reduce rediscovery of already specified mechanics and expose remaining empirical work, but it cannot establish that the transferred contracts improve ORION on fresh tasks.

5 Isolated challenger execution and protected change control

The Shadow development controller composes a governed path of the form

$$\begin{aligned}&\text{persistent issue/evidence} \to \text{research and nearest-work challenge} \to \text{candidate proposal}\\ &\to \text{content-addressed patch} \to \text{isolated sandbox execution} \to \text{regression/fresh assurance}\\ &\to \text{append-only evolution record} \to \text{host recommendation.}\end{aligned}$$

A language model, coding agent, program synthesizer or other worker may generate the proposed change. It is not the controller and does not own the evaluator. The proposed bytes are content-addressed before execution so the evaluated candidate is the candidate later presented for promotion.

Protected evaluator, registry and governance paths are hard rejection coordinates. A candidate that improves visible scores while modifying its own protected evaluation surface is not treated as an improved method. Candidate execution occurs behind a host-owned isolation boundary, and the change controller exposes no merge primitive.

The strongest internally representable successful outcome is therefore a recommendation such as `RECOMMEND_HOST_PROMOTION`, not "self-merged" or "governed by self declaration." This is a structural authority boundary: even a perfect candidate score cannot compensate for missing or compromised change-control evidence.

6 Replay, fresh transfer, and protected assurance

A challenger can overfit the exact failure episodes used to construct it. Self-ORION therefore separates three evidence roles.

Replay. The candidate must address the supporting failure under a bound replay rather than merely changing a proxy score.

Fresh transfer. The candidate must be evaluated on distinct task, split, run/evaluation and variation identities not used as the repair's support set. Improvement on visible development evidence combined with regression on fresh assurance is represented explicitly as `META_OVERFIT` rather than averaged into a favorable mean.

Protected assurance. The authority-bearing evaluator is outside candidate write authority and binds the exact candidate, subject revision, evidence artifact, evaluator identity, lineage, epoch and split. Self-verification, post-hoc evaluator evidence, non-fresh evidence, mixed subject/evaluator epochs and unresolved checks block promotion.

The readiness correction is intentionally asymmetric. Even a complete independent PASS set can reach only `READY_PENDING_EXTERNAL_ATTESTATION`; final promotion remains outside ORION custody. This prevents a boolean checklist, a candidate-generated receipt, or a self-selected benchmark from manufacturing governed Self-ORION authority.

Negative history is part of assurance. A promotion packet that omits failed or harmful alternatives can give a false picture of selection pressure and regression risk; therefore the archive and external handoff must preserve those outcomes.

7 Nearest-work and narrowed novelty boundary

Self-ORION is deliberately not positioned as the first self-editing, self-evolving or issue-persistent agent. ADAS automates agent-system design through a meta-agent and archive [4]; Darwin Gödel Machine performs open-ended self-code modification with empirical validation, an archive and sandboxing [14]; and ADIAS makes persistent issue identity, lifecycle, evidence and intervention-outcome history a first-class optimization state [5]. Persistent issue state is therefore absorbed rather than claimed.

The 2026 failure-recovery literature narrows the residual further. SAGE uses multi-hypothesis failure attribution to generate evidence-grounded explanations and route verified root causes to intervention levels [7]. CausalFlow uses counterfactual intervention to estimate causal responsibility and generate minimal repairs [2]. Failure-driven self-improvement work explicitly converts failed computer-use trajectories into diagnosed code-level improvements [11]. These results mean that failure-driven learning, multi-hypothesis diagnosis and causal attribution are not standalone Self-ORION novelties.

Direct self-evolving coding systems narrow the boundary again. MOSS rewrites agent source from production-failure evidence, validates candidate images by replay and places deployment behind user consent and rollback checks [3]; therefore source-level self-rewriting, replay validation and externally gated deployment are not individually novel. Ratchet manages a self-authored skill library with outcome-driven retirement and bounded capacity while measuring held-out transfer [15]; Verifier-as-Gatekeeper shows that harmful skill admission can contaminate later evolution and tests heterogeneous pre-commit gating plus frozen-pool transfer [9]. MEGA accumulates durable optimization assets in a typed wisdom graph and uses controlled evaluation to attribute improvement effects across optimization cycles [6]. RewardHackingAgents separately makes evaluator tampering and held-out leakage measurable through patch/access telemetry and evaluator locking [1]. Thus lifecycle hygiene, pre-commit gating, durable evolutionary knowledge, effect attribution and evaluator locking are all prior-art pressure rather than sufficient P5 novelty. A current survey of self-evolving coding agents treats feedback reliability, benchmark overfitting, safety, maintainability, cost and generalization as field-level concerns [16], so P5 must measure those concerns rather than present them as unique motivations.

Acceptance-specific work removes another possible novelty route. PACE casts repeated incumbent-versus-candidate commits under adaptive self-evolution as anytime-valid paired sequential testing, so optional-stopping-safe commit acceptance is not a P5 novelty [10]. SEA independently admits self-modifications through anytime-valid certificates around a frozen base [8]. P5 therefore treats these mechanisms as statistical/architectural baseline pressure: an anytime-valid acceptor may be adapted within a stage when noisy paired repeated evaluation requires it, but the surviving candidate is the ordered non-compensatory composition across replay, independent fresh transfer and protected custody rather than the acceptance statistic itself.

PAST-Bench evaluates whether retained experience causes later-task improvement under matched experience-on/off conditions [12]; it motivates pathway-sensitive evaluation rather than treating memory presence as learning. SEVA reports that repeated self-evolution can produce benchmark specialists that improve on one benchmark while regressing on another [13], directly motivating Self-ORION's replay-versus-fresh-transfer and harmful-transfer measurements.

The surviving candidate is therefore narrower and explicitly compositional. A persistent development issue may license a method change only after evidence-bound causal discrimination and invention-readiness checks; acceptance additionally requires non-compensatory staged evidence in the order static checks, replay, independent protected fresh-transfer/non-harm, and protected assurance, with immutable retention of negative/null/harmful evolution history and compromise telemetry, evaluator/holdout custody outside challenger write authority, and structurally absent self-certification or merge authority. Promotion remains a host/external decision. The paper must show that this composition yields more transferable and less compromise-prone improvement than simpler self-edit or acceptance systems; architecture alone cannot establish that claim.

Table 1 provides the full mechanism matrix with dispositions and claim effects. The governed development loop is illustrated in Figure 1.[1]

Table 1: Nearest-work mechanism matrix: 19 mechanisms, dispositions, and their effect on Self-ORION's claim.

Mechanism	Nearest work	What it does	Disp.	Effect on Self-ORION's claim
Persistent issue identity, lifecycle, evidence, intervention-outcome history	ADIAS [5]	First-class issue state across optimization cycles	**ADOPT**	Strikes "persistent issue state" from novelty
Open-ended self-code modification with archive, sandboxing, validation	Darwin Gödel Machine [14]	Self-modifying code with empirical validation; archive and sandbox	**ADAPT**	Strong archive/self-edit/sandbox/validation baseline; P5's distinguishing feature is the attribution-invention-readiness fibre
Meta-agent design search with archive	ADAS [4]	Automates agent-system design through meta-agent and archive	**ADAPT**	Strong meta-agent design-search baseline; P5's fibre is the non-compensatory acceptance ordering
Source rewriting from production-failure evidence, replay validation, externally gated deployment	MOSS [3]	Rewrites agent source from failure evidence; validates by replay; deployment behind user consent and rollback	**ADAPT**	Source-level self-rewriting, replay validation and externally gated deployment are not individually novel
Evaluator-driven program evolution	AlphaEvolve family	Program search driven by evaluator feedback	**ADAPT**	Matched evaluator-only evolutionary-search baseline; P5's claim includes protected custody and fresh transfer
Multi-hypothesis failure attribution with evidence-grounded explanations	SAGE/MHFA [7]	Multi-hypothesis attribution; routes verified root causes to intervention levels	**ADOPT**	Multi-hypothesis failure attribution is prior art; P5's residual is the persistent attribution-invention-readiness cycle
Counterfactual causal attribution and minimal repair	CausalFlow [2]	Counterfactual intervention to estimate causal responsibility; generates minimal repairs	**ADOPT**	Counterfactual causal attribution is prior art; P5's residual is the gated acceptance of the repair over the full evidence chain
Failure-driven trajectory improvement	Failure-driven learning [11]	Converts failed trajectories into diagnosed code-level improvements	**ADOPT**	Failure-driven learning is prior art; P5 must separate "causal attribution" from "gated acceptance"
Skill lifecycle with outcome-driven retirement, bounded capacity, held-out transfer	Ratchet [15]	Self-authored skill library; outcome-driven retirement; bounded capacity; held-out transfer measurement	**ADAPT**	Lifecycle hygiene, capacity management and held-out transfer are prior-art pressure; P5's residual is the non-compensatory evidence ordering and protected custody

[1]The figure is a schematic. Concrete architecture diagrams and pseudocode live in the repository at `research/technical-companions/self-orion-v1/`.

Table P5-T1 (continued): nearest-work mechanism matrix

Mechanism	Nearest work	What it does	Disp.	Effect on Self-ORION's claim
Pre-commit gating, contamination-aware frozen-pool transfer	Verifier-as-Gatekeeper [9]	Harmful skill admission contaminates evolution; tests heterogeneous pre-commit gating and frozen-pool transfer	**ADOPT**	Pre-commit gating and contamination-aware frozen-pool transfer are prior art; P5 relies on narrower authority/causal/integrity composition
Evaluator tampering, holdout leakage, access telemetry, evaluator locking	RewardHackingAgents [1]	Makes evaluator tampering and holdout leakage measurable through patch/access telemetry and evaluator locking	**ADOPT**	Evaluator integrity mechanics are explicit benchmark parents; P5 must show the composition yields better outcomes
Durable optimization assets, typed wisdom, controlled effect attribution	MEGA [6]	Accumulates durable optimization assets in a typed wisdom graph; controlled evaluation for effect attribution	**ADAPT**	Durable evolutionary knowledge and effect attribution are prior-art pressure; P5's residual is the evidence-bounded acceptance chain
Unrestricted self-evolving coding agents survey	Current frontier [16]	Surveys feedback reliability, benchmark overfitting, safety, maintainability, cost, generalization	**ADAPT**	Field-level concerns are documented; P5 must measure them, not present them as unique motivations
Anytime-valid paired sequential commit acceptance	PACE [10]	Incumbent-vs-candidate commits under adaptive self-evolution as anytime-valid paired sequential testing	**ADAPT**	Optional-stopping-safe commit acceptance is not a P5 novelty; may be adapted as within-stage statistical acceptor
Anytime-valid self-modification certificates	SEA [8]	Admits self-modifications through anytime-valid certificates around a frozen base	**ADAPT**	Anytime-valid certificate architecture pressures claims around statistically certified self-modification acceptance
Pathway-sensitive experience-on/off fresh-transfer evaluation	PAST-Bench [12]	Evaluates whether retained experience causes later-task improvement under matched experience-on/off conditions	**ADOPT**	Pathway-sensitive evaluation is prior art; motivates P5's replay-vs-fresh decomposition
Specialist regression and harmful transfer measurement	SEVA [13]	Repeated self-evolution produces benchmark specialists that improve on one benchmark while regressing on another	**ADOPT**	Makes harmful fresh transfer a first-class outcome; directly motivates P5's replay-vs-fresh-transfer and harmful-transfer measurements
Experience retention and replay	Continual experiential learning (parent discipline)	Retains and replays past experience for later reuse	**COMPOSE**	Experience retention/replay is not standalone novelty; P5's claim is the protected ordering of replay, fresh transfer and host custody
Attribution/repair quality measurement	Debugging and root-cause program repair (parent disciplines)	Locates and repairs defects using diagnostic evidence	**COMPOSE**	Measure attribution/repair quality; do not relabel debugging as novelty
Protected custody, contamination telemetry, tail harms as non-compensatory gates	Governance parent disciplines and self-evolving-agent safety work	Models controlled evolution as a safety/authority problem with non-compensatory gates	**COMPOSE**	Protected custody, contamination telemetry and tail harms are non-compensatory gates; P5 must close on the composition

Dispositions: **ADOPT** 8 (mechanism absorbed; stripped from novelty) · **ADAPT** 8 (mechanism absorbed with a stated variant) · **COMPOSE** 3 (constrains the composition, not a parent) · **DEFER** 0 · **REJECT** 0. Row total: 19.

8 Falsifiers and empirical programme

8.1 Local hostile falsifier V1

The current local suite exercises the governance semantics rather than claiming open-world improvement. Registered cases include recurrence without causality, competing-cause discrimination, negative intervention retention, replay/fresh-transfer separation, evidence-bound reopening, invention-readiness constraints and protected-path reward-hacking attacks. A candidate that modifies protected registry/governance state is rejected even when visible and fresh score deltas are constructed to look perfect.

These tests support the statement that the implementation can represent and refuse the registered failure modes. They do not show that Self-ORION discovers correct causes or produces beneficial repairs on real development problems.

8.2 Prospective external comparison

The outcome-blind design is frozen as `P5.hidden-cause-fresh-transfer.v1`. The primary hypothesis is that full Self-ORION produces greater *protected fresh-task improvement* than the strongest matched self-edit/self-evolution baseline on hidden-cause development cases, while harmful-transfer rate remains within a predeclared two-percentage-point non-inferiority guard. The design target for fresh improvement is an absolute five-percentage-point advantage. These values are prospective margins, not observed effects.

Registered hidden cause families are retrieval miss, routing/planning miss, implementation bug, environment/dependency/tool failure, evaluator/metric bug, representation gap, measurement/specification gap and genuine method-basis gap. The candidate sees the symptom/context but not the protected root-cause label. Each episode separates motivating evidence, replay tasks and an independently frozen fresh-transfer set that changes at least one task/domain/model/environment/data/tool axis.

Strong baselines include a fixed agent, direct self-edit, ADAS/meta-agent search, DGM-like archive/self-edit, ADIAS-like issue-centric optimization, SAGE/MHFA-like multi-hypothesis attribution, CausalFlow-like counterfactual repair, failure-driven patching and evaluator-only evolutionary search where runnable. Ablations remove persistent issue state, causal discrimination, invention readiness, independent fresh transfer, protected evaluator custody, negative history and access telemetry; a self-certification/promotion ablation is permitted only in an isolated benchmark safety lane and cannot affect repository/main authority.

Primary outcomes are protected fresh-task improvement, fresh-transfer success and harmful-transfer rate. Secondary outcomes include cause macro-F1, false method-change rate, intervention success conditional on attribution, evaluator/holdout/negative-history compromise, correct block/`CANNOT_CHECK`, recognized-failure recurrence, negative-history completeness, motivating-task improvement and resource cost. Wilson intervals are used for standalone rates and paired percentile bootstrap intervals for matched improvement differences; stochastic repetitions are nested within episode/system. Harmful tail/family regressions are reported separately rather than averaged away.

8.3 Prospective design freeze and fresh-transfer custody

The complete protocol lives at `protocol/PROTOCOL_V1.json`; case structure is frozen in `protocol/HIDDEN_CAUSE_C`
and `protocol/FRESH_TRANSFER_POLICY_V1.md` defines replay/freshness and harm accounting. The current state is `DESIGN_FROZEN` with `outcome_accessed=false`.

This is not yet an execution freeze. The exact subject revision, hidden-cause suite hash, motivating/replay/fresh split hashes, provider/model versions, baseline configs, evaluator artifact and evaluation epoch remain unbound. The fresh evaluator/holdout must be frozen outside challenger write authority before candidate generation, and the fresh set cannot be recycled as development feedback without losing freshness. Search/file/evaluator/patch access is retained for contamination and integrity review. PAST-Bench is reference-pinned to `Gen-Verse/PAST-Bench@f8223517ae7491e776b69793d9f11e9d` any actual use still requires a frozen task/split manifest.

8.4 Statistical methods

Standalone rates use Wilson 95% intervals. Matched improvement differences, when an archive exists, use paired percentile bootstrap with 10 000 resamples nested in episode × system × seed. The primary H1 family is protected fresh-task improvement with a harmful-transfer guard; Holm correction applies to inferential secondary attribution/governance families. Catastrophic fresh regressions are reported by family and are not averaged away. Missing protected evidence remains `CANNOT_CHECK`, never zero. The diagnostic 21/24 attribution interval in Table 2 is a standalone Wilson interval on $n = 24$; it is not an H1 effect size.

8.5 Threat model

The evaluation treats evaluator compromise, holdout leakage, search contamination, replay-only laundering, missing-stage compensation, negative-history deletion and self-certification as first-class failure modes rather than as missing data. A candidate that can rewrite the protected evaluator, recycle the fresh set as development feedback, or certify its own readiness has not produced an admissible improvement. Local hostile tests exercise registered instances of those attacks; unknown gaming strategies remain outside the present evidence.

8.6 External promotion boundary

The external governed self-improvement state remains `CANNOT_CHECK` until the execution identities are bound and the prospective observations exist. Repository CI, structural specification closure, local hostile tests, replay-only success, a diagnostic 21/24 attribution score, or an internal readiness state cannot promote that boundary. Even a positive fresh-transfer result is evidence for an external host recommendation rather than self-merge authority.

9 Diagnostic attribution archive

The only result-bearing P5 archive currently bound in the repository is a single-model glm-5.2 hidden-cause attribution run over 24 cases (three cases in each of the eight registered families). Metrics below are recomputed from the raw JSONL; the sidecar report is checked against those rows and is refused if it disagrees.

Accuracy is $21/24 = 0.875$ (Wilson 95% interval in Table 2). This is a descriptive diagnostic score. It is not protected fresh-task improvement, not a matched baseline comparison, and not an H1 result.

Three residual errors are retained and are not relabeled as successes (Table 3):

- `P5-HC-002`: gold `RETRIEVAL_MISS`, attributed `REPRESENTATION_GAP` (MEDIUM);

Table 2: Hidden-cause attribution confusion matrix from the archived glm-5.2 diagnostic run. Accuracy is 21/24 (Wilson 95% [0.690, 0.957]). Three residual errors are retained and are not relabeled as successes. This is not a protected fresh-transfer or H1 result.

gold / attr.	Retr	Rout	Impl	Env	Eval	Repr	Meas	Meth
Retr	2	0	0	0	0	1	0	0
Rout	0	3	0	0	0	0	0	0
Impl	0	0	3	0	0	0	0	0
Env	0	0	1	2	0	0	0	0
Eval	0	0	0	0	3	0	0	0
Repr	0	0	0	0	0	2	0	1
Meas	0	0	0	0	0	0	3	0
Meth	0	0	0	0	0	0	0	3

Residual errors: P5-HC-002: gold RETRIEVAL_MISS → REPRESENTATION_GAP (MEDIUM) P5-HC-012: gold ENVIRONMENT_DEPENDENCY_TOOL_FAILURE → IMPLEMENTATION_BUG (HIGH) P5-HC-018: gold REPRESENTATION_GAP → METHOD_BASIS_GAP (HIGH).

Table 3: Retained residual attribution errors from the archived glm-5.2 diagnostic run. These three cases remain incorrect; they are not successes.

case	gold	attributed	confidence
P5-HC-002	RETRIEVAL_MISS	REPRESENTATION_GAP	MEDIUM
P5-HC-012	ENVIRONMENT_DEPENDENCY_TOOL_FAILURE	IMPLEMENTATION_BUG	HIGH
P5-HC-018	REPRESENTATION_GAP	METHOD_BASIS_GAP	HIGH

- `P5-HC-012`: gold `ENVIRONMENT_DEPENDENCY_TOOL_FAILURE`, attributed `IMPLEMENTATION_BUG` (HIGH);

- `P5-HC-018`: gold `REPRESENTATION_GAP`, attributed `METHOD_BASIS_GAP` (HIGH).

Replay-versus-fresh scatter, longitudinal specialist-regression, improvement/integrity frontier, failure-recurrence, cost-to-protected-improvement, baseline/ablation Table P5-T2, and campaign harmful/null Table P5-T3 have no admissible rows in this archive. Those artifacts are emitted as `CANNOT_CHECK` stubs by `python -m orion.study.p5.tables` and are not imputed from the 21/24 accuracy.

10 Limitations and threats to validity

The present revision establishes an implemented Self-ORION governance fibre, a local hostile falsifier, a frozen prospective protocol, and one diagnostic attribution archive. It does not establish transferable protected improvement, reduced harmful transfer, or integrity advantage against matched self-edit/self-evolution baselines.

Diagnostic attribution is not H1. The glm-5.2 archive scores 21/24 on hidden-cause labels. That quantity is a single-model, single-run descriptive accuracy with a wide Wilson interval. It does not measure protected fresh-task improvement, harmful transfer, or false protected acceptance. The three residual errors remain errors.

Table 4: Replay-vs-fresh scatter: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no campaign-level replay/fresh score pairs. This table will be populated when a live protected-transfer campaign produces matched replay and fresh-task measurements.

Round	Replay score	Fresh score
`CANNOT_CHECK` — no admissible data in archive		

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 5: Longitudinal specialist regression: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no longitudinal specialist-designation data. This table will be populated when a live campaign produces round-by-round specialist scores with regression annotations.

Round	Specialist	Prior score	Current score
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

The 24-case suite is not an execution-frozen campaign. The repository suite covers eight families, but evaluator hashes and several fresh-task hashes remain placeholders. Motivating/replay/fresh split identities, provider/model revisions, and the protected evaluator epoch are `UNBOUND` in `PROTOCOL_V1.json` and `PROTOCOL_V2.json`.

No matched baselines or ablations were executed. Registered DGM/ADAS/ADIAS/SAGE/CausalFlow/P comparators and governance ablations exist as protocol names. They have no result-bearing archive on this subject, so Table P5-T2 is `CANNOT_CHECK`.

No live #8/#76 cycle is bound. The live-trial packet still records `corpus_revision: UNBOUND`. Provider and protected-verifier credentials were unset in this session, so no new live campaign was started. A previously advertised perfect-score report is not used: its artifact identity and git identity diverged before integration.

Statistical threats. $n = 24$ with one seed cannot support the predeclared H1 five-percentage-point fresh-improvement target or the two-percentage-point harm guard. There is no paired bootstrap against a baseline, no multiplicity-adjusted secondary family, and no tail/family regression from a campaign. Macro-F1 in the original sidecar assumed perfect precision and is not promoted.

Contamination and evaluator custody. The diagnostic run used locally visible gold labels to score attributions after the fact. That is admissible for an offline label-recovery diagnostic. It is not admissible as protected-evaluator evidence: the candidate-facing study still requires labels, fresh payloads and scoring rubrics to remain outside challenger custody.

Table 6: Improvement-integrity frontier: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no frontier-pair data. This table will be populated when a live campaign produces matched improvement and integrity scores across rounds.

Round	Improvement score	Integrity score
`CANNOT_CHECK` — no admissible data in archive		

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 7: Failure recurrence: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no recurrence annotations. This table will be populated when a live campaign records which previously resolved failures recur across rounds.

Original failure	Round resolved	Round recurred	Outcome
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

11 Ethics, safety, and reproducibility

11.1 Ethical considerations

This paper studies governed method evolution for research software. The frozen local falsifier and the diagnostic attribution suite use constructed hidden-cause cases. They do not involve human-subject records or personally identifiable information. No live provider campaign was executed in this revision.

11.2 Safety and authority

Self-ORION has no self-merge primitive. The strongest internal terminal remains a host-only promotion recommendation. Isolated self-certification ablations, if ever executed, are forbidden on the production/main authority path. Negative, null, blocked and cannot-check outcomes are retained rather than dropped.

11.3 Reproducibility and data/code availability

Source, protocol, the diagnostic JSONL, and the table generator are version-controlled in `SzeChunYiu/ORION`. Persistent DOI assignment remains pending publication.

Archived-result regeneration, without executing a live system, is:

```
make paper05-results
```

which runs `python -m orion.study.p5.tables`. The command recomputes 21/24 from `evidence/glm-5.2-a` writes Table P5-3 and the residual-error ledger, and writes `CANNOT_CHECK` stubs for P5-2, P5-4, P5-5, P5-6, P5-7 and Tables P5-T2/P5-T3. A rewrite of the archive that hides the three errors is refused (exit 2). A missing archive is `CANNOT_CHECK` (exit 3). Exact commands are in `papers/paper-05-self-orion/REPRODUCE.md`.

Table 8: Cost to validated improvement: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no cost-to-improvement annotations. This table will be populated when a live campaign records the resource cost (e.g., API calls, rounds, wall time) required to reach validated improvement.

Improvement	Validation	Cost metric	Value
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 9: Baseline and ablation transfer/integrity results: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no baseline/ablation arm data. This table will be populated when a live campaign produces matched baseline-vs-ablation transfer and integrity measurements.

Condition	Transfer score	Integrity score	Δ vs. baseline
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

11.4 Compute and conservative-abstention cost

The diagnostic archive records 14 190 tokens and mean latency ≈13 s per case for glm-5.2 attribution only. Cost and time to *protected validated improvement* remain `CANNOT_CHECK`. No campaign-level abstention/cost trade-off is reported.

11.5 Licensing

This revision does not add a repository-level license. Redistribution terms remain `CANNOT_CHECK` from the repository state until they are declared explicitly.

12 Nearest-work and claim boundary

Self-ORION does not claim the first self-improving agent, the first evolutionary program search system, the first agent archive, the first issue-centric self-improvement state, or the first failure-driven adaptation loop. The programme explicitly absorbs mechanisms from ADAS, Darwin Gödel Machine, A Self-Improving Coding Agent, AlphaEvolve, AI Co-Scientist evolution, ADIAS-style persistent issue state and evaluation-integrity research.

The narrower candidate comparison is whether a failure-governed method-evolution fibre with persistent negative history, causal discrimination, invention-readiness gating, isolated challenger execution, replay, fresh transfer, protected assurance and no self-promotion provides a safer and more reliable development process than simpler self-edit/evolution baselines. Method-level challenger governance is a P5 adoption boundary only: it does not claim that P10 can learn an innovation grammar or invent useful methods.

Prospective V3 work now adds a narrower formal safeguard before any broader self-revision experiment: competing revision-responsibility hypotheses may remain ambiguous or unresolved, and required interface-adequacy checks may block escalation from an observed failure to a model, method or objective rewrite while a narrower observation/measurement/representation repair re-

Table 10: Campaign harmful/null interventions: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no intervention-level outcome annotations. This table will be populated when a live campaign records which interventions were harmful, null, or beneficial relative to the incumbent.

Intervention	Outcome	Effect size	Recovery
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

mains indicated. The implementation can nominate a claim-relative minimal candidate mechanic for later testing, but the responsibility report, interface report and revision gate are all structurally non-authorizing. They do not replace the current P5 invention-readiness gate, do not grant adoption or promotion, and do not supply evidence for H1–H4. Their scientific value remains a prospective question under the revision-level discrimination study.

A second successor tranche makes three additional distinctions explicit without granting new operating authority. First, internal computation is itself represented as a costed epistemic action: hard verification obligations can require an expensive check even when a cheaper LLM/planning action has greater scalar expected value, while optional non-positive computation may stop locally. Second, an uncertainty-containment envelope can restrict the contexts in which a model or claim is admissible without rewriting the model or certifying it as correct. Third, social corroboration is evidence-bound: different worker/verifier identities are not counted as independent when their registered upstream sources or direct observations overlap, and unresolved hidden contributors prevent exclusive causal credit. These are implementation contracts for later countermodels and experiments, not evidence that adaptive computation, containment or social-state tracking improves Self-ORION.

The prospective V3 experiment is now implemented as a separate, protected-custody revision-level benchmark rather than as a rewrite of the frozen P5 V1 protocol. Its development-only cause-confusable panel exposes the same candidate-visible symptom with different protected revision responsibilities, evaluates matched no-revision, broad-self-edit, M-open-only, world-model, representation-regime, generic-diagnosis and full T7 policies, and includes feedback permutation/no-feedback controls. A separate scorer evaluates protected revision responsibility, false broad revision, unresolved decisions and later fresh-transfer/harm records. This establishes only that the experiment can be run and audited. The confirmatory preflight remains `CANNOT_CHECK`: final subject/split/evaluator/protected-suite identities are unbound and the load-bearing donor comparators do not yet have frozen structural bindings from the #454 assimilation programme. The development panel therefore cannot support H1–H4, close the V3 study, or make P5 peer-review ready.

Several parts of that statement are already implementation contracts; the performance claim is not established. In particular, architecture readiness does not imply that candidate causes are scientifically correct, that the development policy chooses valuable work, that repairs transfer, or that the protected evaluator is sufficient against unknown gaming strategies.

The former Shadow Mechanics manuscript is therefore not a sixth scientific contribution. Its mechanic representation belongs to Paper I, its failure-to-method and self-driving development theory belongs here, and its historical manuscript/evidence packet is retained only as a technical provenance archive under `research/technical-companions/mechanics-of-mechanics-v1/`.

13 Conclusion

Self-ORION treats method evolution as a research-and-authority problem rather than an invitation for an agent to edit itself. Failure episodes remain evidence, recurrence remains distinct from cause, issues persist across discriminators and interventions, invention is gated by search and attribution, challengers execute in isolation, replay is separated from fresh transfer, protected assurance is external to candidate write authority, and final promotion remains in host custody. The V3 successor formalism sharpens the same boundary: an anomaly is not itself permission to revise, an unresolved responsibility is not a license to choose among incompatible repairs, an interface defect is not evidence that the method basis should be rewritten, more expected computational value is not scientific authority, a validity region is not a truth certificate, and repeated social reports are not independent evidence merely because they came from different processes.

The local hostile programme demonstrates these registered governance semantics and has already exposed authority failures in earlier readiness designs. The diagnostic glm-5.2 archive scores 21/24 on hidden-cause labels and retains three residual errors; that descriptive result does not support H1–H4. The V3 revision-level experiment is now prospectively implemented but not confirmatory-frozen or executed, so it also supplies no positive scientific result. The scientific question remains whether this architecture improves fresh development outcomes while reducing harmful repair, evaluator gaming, false broad revision and false promotion relative to matched alternatives. Until the original P5 campaign or a separately frozen V3 campaign executes under complete protected identities and independent verification, governed Self-ORION remains `CANNOT_CHECK`. Claim-to-evidence bindings for this revision live in `evidence/CLAIM_LEDGER_V1.md`.

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